

LUCAS MOL

Senior Software Engineer | Backend & Distributed Systems

Niterói, Rio de Janeiro, Brazil | **Open to remote (US/CA time zones) & relocation**

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PROFESSIONAL SUMMARY

Senior Software Engineer with 7+ years building and modernizing backend systems in Java and Go. Experienced in high-availability microservices, asynchronous and event-driven architectures, performance optimization and observability for mission-critical, high-traffic platforms. Led a development team through a legacy monolith-to-microservices migration. Pursuing an MBA in Software Architecture and applying DevSecOps practices across the software lifecycle.

TECHNICAL SKILLS

Languages: Java (8 / 17 / 21), Go, JavaScript, TypeScript, Python

Frameworks & Libraries: Spring Boot, Spring Framework, Hibernate, Java EE (EJB, JSF), Node.js, NestJS, Express, Prisma, TypeORM

Databases & Messaging: PostgreSQL, MongoDB, MS SQL Server, Redis, Kafka, RabbitMQ

AWS: EC2, S3, Lambda, SQS, EventBridge, Step Functions, End User Messaging (SNS), SSM, Secrets Manager

DevOps & Observability: Docker, Kubernetes, Helm, Rancher, Terraform, CI/CD (Jenkins, ArgoCD), Datadog, Splunk, Grafana, Git, Maven

Practices: Microservices, Event-Driven Architecture, DDD, TDD, Clean Code, Design Patterns, RESTful APIs (OpenAPI/Swagger)

PROFESSIONAL EXPERIENCE

Senior Software Engineer

Dock — Brazil (Remote) | Nov 2025 - Present

B2B Banking-as-a-Service platform serving hundreds of companies.

- Build and operate 24/7 multi-tenant microservices (Java, Spring Boot, Kafka, AWS) processing 20M+ transactions per day across instant payments (PIX), cards and invoicing.
- Implemented a Deadline Propagation strategy across service layers, ensuring consistent rollbacks for timed-out operations.
- Migrated third-party integrations with critical latency issues to asynchronous processing, reducing timeout occurrences and cutting p99 latency from ~15s to ~3s.
- Designed a centralized keyset pagination solution, cutting query time by 40% beyond page 5,000, with gains growing with depth.
- Standardized request context and metrics/tracing exposure (Datadog, Splunk, Grafana), improving incident response.
- Deploy and maintain services on Kubernetes and Terraform-managed AWS infrastructure alongside SRE teams, using Helm per environment with Rancher, Jenkins and ArgoCD, and progressive rollouts through partial cluster-by-cluster and canary deployments.

Tech Lead / Senior Software Developer

CAPGov - COPPETEC — Rio de Janeiro, Brazil (Remote) | Jul 2024 - Oct 2025

Research and technology foundation of the Federal University of Rio de Janeiro (UFRJ).

- Led a team of 4 developers on technical initiatives for the Rio de Janeiro Public Defender's Office, with systems used by hundreds of staff and thousands of citizens.
- Spearheaded the end-to-end modernization of a Java 8 JAX-RS monolith into 4 Spring Boot microservices and Go schedulers.
- Owned critical cross-project initiatives, including AWS and third-party integrations and OIDC/OAuth 2.0 single sign-on across multiple services (RESTful APIs, full-stack JSF applications and others), unifying authentication across the organization's applications.

Software Developer

CAPGov - COPPETEC — Rio de Janeiro, Brazil (Remote) | Nov 2021 - Jul 2024

- Developed and maintained high-availability RESTful APIs and full-stack JSF applications with Enterprise Java (EJB), using TDD and DDD with strong test coverage.
- Reduced average response time by 88.5% and improved peak-load performance 6.9x under stress tests through optimized database queries and concurrency routines.

Software Development Intern

Centro Universitário UniCarioca — Rio de Janeiro, Brazil | Aug 2018 - Feb 2020

- Developed a facial-recognition authentication prototype for images and video using OpenCV and machine learning.
- Built an application to assess student learning levels in early childhood education, and web applications in Java and C# supporting master's research projects.

EDUCATION

MBA in Software Architecture

Faculdade Full Cycle de Tecnologia | In Progress, expected 2028

Bachelor's Degree in Computer Science

Centro Universitário Carioca — Rio de Janeiro, Brazil | Jul 2018 - Dec 2023

LANGUAGES

Portuguese (Native) | English (Advanced - technical proficiency)